

Modify The Matrix

Size = 8×10

arr

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	0	1	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	1	0	0
4	0	0	0	0	0	0	0	0	0	0
5	0	0	0	0	0	0	0	0	0	0
6	0	0	0	0	0	0	0	0	0	0
7	0	0	0	0	1	0	0	0	0	0

approach 1

by creating a copy of original
2d array

↳ traversing in 1 array and
updating in another

$$T.C = O(m * n), \quad \underline{\underline{S.C = O(m * n)}}$$

Approach 2

cols

0	0	1	0	1	0	0	1	0	0
0	1	2	3	4	5	6	7	8	9

rows

0	0
1	1
2	0
3	1
4	0
5	0
6	0
7	1

0	0	0	1	0	1	0	0	1	0	0
1	1	1	1	1	1	1	1	1	1	1
2	0	0	1	0	1	0	0	1	0	0
3	1	1	1	1	1	1	1	1	1	1
4	0	0	1	0	1	0	0	1	0	0
5	0	0	1	0	1	0	0	1	0	0
6	0	0	1	0	1	0	0	1	0	0
7	1	1	1	1	1	1	1	1	1	1

i j
 ↑ ↑
 (1, 2)
 (3, 7)
 (7, 4)

$$T.C = O(m \times n), \quad S.C = O(m + n)$$

pseudo
code

i/p 2D array ($m \times n$)

1) create 1d array of size m // rows

2) create 1d array of size n // cols

3) traverse in 2d array

3.1) check if $arr[i][j] == 1$

update $rows[i] = 1$

$cols[j] = 1$

4) traverse in 2d array

4.1) check if $row[i] == 1$ or $cols[j] == 1$

update $arr[i][j] = 1$

return 2d array

code

S.C = $O(m+n)$, T.C = $O(m*n)$

```
public static int[][] modifyMatrix(int[][] arr, int m, int n) {  
    int[] rows = new int[m];  
    int[] cols = new int[n];  
    for (int i = 0; i < m; i++) {  
        for (int j = 0; j < n; j++) {  
            if (arr[i][j] == 1) {  
                rows[i] = 1;  
                cols[j] = 1;  
            }  
        }  
    }  
    for (int i = 0; i < m; i++) {  
        for (int j = 0; j < n; j++) {  
            if (rows[i] == 1 || cols[j] == 1) {  
                arr[i][j] = 1;  
            }  
        }  
    }  
    return arr;  
}
```

⇒ Strings

```
str = "A b c d e f g" ;
```

inbuilt

- 1) `str.length()`; // 7
- 2) `str.charAt(index)`; // 'd'

- 3) `str.toUpperCase()`; // "ABCDEFGH"
- 4) `str.toLowerCase()`; // "abcdefgh"

`str = "ABCDEFG" ;`
0 1 2 3 4 5 6

substring :- sub part of a string

Syntax 1

`str.substring ( start index, end index + 1 );`

- 1) `str.substring (1, 5);` // bcde
- 1) `str.substring (0, 5);` // abcde
- 1) `str.substring (3, 5);` // de
- 1) `str.substring (3, 4);` // d
- 1) `str.substring (4, 6);` // ef
- 1) `str.substring (4, 7);` // efg
- 1) `str.substring (4, 8);` // error

Syntax 2

str = "ABCDEFGH"
0 1 2 3 4 5 6

str.substring(start index);

str.substring(4); // EFG

Print Characters

```
public static void main(String[] args) {  
    Scanner scn = new Scanner(System.in);  
    String str = scn.nextLine();  
  
    for (int i = 0; i < str.length(); i++) {  
        System.out.println( str.charAt(i) );  
    }  
}
```

Is Equal?

observation

↳ length of both strings should be same

↳ each char at same index should also be same

code

$T.C = O(\text{str1.length()})$

```
public static void main(String[] args) {
    Scanner scn = new Scanner(System.in);
    String str1 = scn.nextLine();
    String str2 = scn.nextLine();

    System.out.println(checkEqual(str1, str2));
}

public static boolean checkEqual(String str1, String str2) {
    if ( str1.length() != str2.length() ) {
        return false;
    } else {
        for (int i = 0; i < str1.length(); i++) {
            if ( str1.charAt(i) != str2.charAt(i) ) {
                return false;
            }
        }
        return true;
    }
}
```

Note:-

strings can't be compared

with ==

instead use str1.equals(str2);

code

```
public static void main(String[] args) {  
    Scanner scn = new Scanner(System.in);  
    String str1 = scn.nextLine();  
    String str2 = scn.nextLine();  
  
    boolean ans = str1.equals(str2);  
  
    System.out.println(ans);  
}
```

Print Indices of Vowels

str = "KunalSuri";
0 1 2 3 4 5 6 7 8

[o/p] 1 3 6 8

```
public static void main(String[] args) {  
    Scanner scn = new Scanner(System.in);  
    String str = scn.nextLine();  
  
    for (int i = 0; i < str.length(); i++) {  
        char ch = str.charAt(i);  
        if (isVowel(ch)) {  
            System.out.print(i + " ");  
        }  
    }  
}  
  
public static boolean isVowel(char ch) {  
    return ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' || ch == 'A' || ch == 'E' ||  
    ch == 'I' || ch == 'O' || ch == 'U';  
}
```